

MAT 324 HW11

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Due 12/04

Problem 1. (a) (Axler 11B #2) Use the result of Exercise 12(a) in Section 11A to show that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots = \frac{\pi^2}{8}$$

(b) Deduce the same identity from the identity in Axler's 11.32.

Solution. (a) Let f be as in Exercise 11A #12(a). Since $|f|^2 = 1$ a.e., $\|f\|_2^2 = 1$. By Parseval's identity,

$$1 = \|f\|_2^2 = \sum_{n \in \mathbb{Z}} |\widehat{f}(n)|^2 = 2 \sum_{n=1}^{\infty} \frac{4}{(2n-1)^2 \pi^2} = \frac{8}{\pi^2} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots \right),$$

which implies what we want.

(b) We can subtract the even terms away from Axler's 11.32:

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \left(1 - \frac{1}{4}\right) \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{3}{4} \cdot \frac{\pi^2}{6} = \boxed{\frac{\pi^2}{8}}.$$

There is no cause for concern with the first equality because all the sums above converge absolutely. □

Problem 2. Use Green's, Stokes', or residue theorem to show that

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y}{x^2 + y^2} e^{-2\pi i t x} dx = e^{-2\pi y |t|}$$

for all $y \in \mathbb{R}^+$ and $t \in \mathbb{R}^*$.

Solution. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be

$$f(z) = \frac{1}{\pi} \cdot \frac{y}{z^2 + y^2} e^{-2\pi i t z},$$

noting that f has two singularities at $z = \pm iy$. Assume $t \leq 0$, so that $e^{-2\pi i t z}$ is decreasing with the imaginary part of z . Let $R > y$ and let $\gamma_R \subseteq \mathbb{C}$ be the boundary of the region $\{z \in \mathbb{C} : |z| \leq R \text{ and } \operatorname{Im} z \geq 0\}$, oriented positively. Since $|f(z)| < \frac{M}{|z|^2}$ for some constant M and sufficiently large $|z|$ (with nonnegative imaginary part), the integral over the semicircle becomes arbitrarily small as $R \rightarrow \infty$, so

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \oint_{\gamma_R} f(z) dz.$$

Since iy is a simple pole of f , the residue of f at iy can be computed as

$$\lim_{z \rightarrow iy} (z - iy) f(z) = \lim_{z \rightarrow iy} \frac{1}{\pi} \cdot \frac{y(z - iy)}{(z + iy)(z - iy)} e^{-2\pi i t z} = \lim_{z \rightarrow iy} \frac{1}{\pi} \cdot \frac{y}{z + iy} e^{-2\pi i t z} = \frac{1}{2\pi i} e^{2\pi t y} = \frac{1}{2\pi i} e^{-2\pi |t| y}$$

where the last equality is because $t \leq 0$. By the residue theorem,

$$\oint_{\gamma_R} f(z) dz = 2\pi i \cdot \frac{1}{2\pi i} e^{-2\pi |t| y},$$

implying the result. Similarly, if $t \geq 0$, then we instead use residue theorem on γ'_R , the reflection of γ_R over the real axis, in the exact same fashion. □

Problem 3. (a) Let $(X, \mathcal{S}, \mu)$ be a measure space, $n \in \mathbb{N}$, $p_1, \dots, p_n, r \in (0, \infty]$ with $1/p_1 + \dots + 1/p_n = 1/r$, and $f_1, \dots, f_n: X \rightarrow [0, \infty]$ be measurable functions. Show that

$$\|f_1 \cdots f_n\|_r \leq \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}.$$

(b) Let $p, q, r \in [1, \infty)$ with $1/p + 1/q - 1 = 1/r$ and $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be measurable functions. Show that for all $x \in \mathbb{R}$,

$$|(f * g)(x)|^r \leq \|f\|_p^{r-p} \|g\|_q^{r-q} \int_{\mathbb{R}} |f(y)|^p |g(x-y)|^q d\lambda(y).$$

(c) Let $p, q, r \in [1, \infty]$ with $1/p + 1/q - 1 = 1/r$ and $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be measurable functions. Show that

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

Solution. (a) We prove the inequality by induction. The base case $n = 2$ is the result of HW09 #1(a). Suppose that the statement holds for $n - 1$ functions f_i and numbers p_i , and let $x \in (0, \infty]$ be such that $1/p_2 + 1/p_3 + \dots + 1/p_n = 1/x$. Since $1/p_1 + 1/x = 1/r$, the statement for 2 functions and for $n - 1$ functions together gives

$$\|f_1 \cdots f_n\|_r \leq \|f_1\|_{p_1} \|f_2 \cdots f_n\|_x \leq \|f_1\|_{p_1} \cdot \|f_2\|_{p_2} \cdots \|f_n\|_{p_n}.$$

(b) Given $x \in \mathbb{R}$, define $g_x: \mathbb{R} \rightarrow \mathbb{R}$ by $g_x(t) = g(x - t)$ for all $t \in \mathbb{R}$. Then, taking both sides to the $1/r$ power, the desired inequality can be rewritten as

$$\|fg_x\|_1 \leq \|f\|^{1-p/r} \|g\|_{pr/(r-p)} \cdot \|g_x\|^{1-q/r} \|g\|_{qr/(r-q)} \cdot \|f\|^{p/r} \|g_x\|^{q/r} \|r,$$

which holds by part (a), noting that we are allowed to apply (a) since

$$\left(\frac{pr}{r-p}\right)^{-1} + \left(\frac{qr}{r-q}\right)^{-1} + r^{-1} = \left(\frac{1}{p} - \frac{1}{r}\right) + \left(\frac{1}{q} - \frac{1}{r}\right) + \frac{1}{r} = \frac{1}{p} + \frac{1}{q} - \frac{1}{r} \stackrel{?}{=} 1$$

by the given relation on p, q , and r .

(c) Integrate the inequality proven in (b) and factor out the constants on the right-hand side to get

$$\|f * g\|_r^r \leq \|f\|_p^{r-p} \|g\|_q^{r-q} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(y)|^p |g(x-y)|^q d\lambda(y) \right) d\lambda(x).$$

The double integral on the right-hand side evaluates to

$$\begin{aligned} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(y)|^p |g(x-y)|^q d\lambda(y) \right) d\lambda(x) &= \int_{\mathbb{R}} |f(y)|^p \left(\int_{\mathbb{R}} |g(x-y)|^q d\lambda(x) \right) d\lambda(y) \\ &= \int_{\mathbb{R}} |f(y)|^p d\lambda(y) \int_{\mathbb{R}} |g(x)|^q d\lambda(x) = \|f\|_p^p \|g\|_q^q, \end{aligned}$$

where the first equality is justified by Tonelli's theorem, as the integrand is nonnegative. Substituting this into the inequality gives

$$\|f * g\|_r^r \leq \|f\|_p^r \|g\|_q^r,$$

which is equivalent to the desired result. $\square$

Problem 4. (a) Suppose $f, \widehat{f} \in L^1(\mathbb{R}; \mathbb{C})$. Show that $f, \widehat{f} \in L^2(\mathbb{R}; \mathbb{C})$.

(b) Suppose $f \in L^1(\mathbb{R}; \mathbb{C})$ is differentiable with $f' \in L^1(\mathbb{R}; \mathbb{C})$. Show that $\widehat{f} \in L^2(\mathbb{R}; \mathbb{C})$.

(c) Suppose $f \in L^1(\mathbb{R}; \mathbb{C})$ is twice continuously differentiable with $f', f'' \in L^1(\mathbb{R}; \mathbb{C})$. Show that $\widehat{f} \in L^1(\mathbb{R}; \mathbb{C})$.

Solution. (a) By the Axler's 11.49, $\widehat{\widehat{f}}, \widehat{\widehat{f}} \in L^\infty(\mathbb{R}; \mathbb{C})$, and $\widehat{\widehat{f}}(x) = f(-x)$ for all $x \in \mathbb{R}$, so $f \in L^\infty(\mathbb{R}; \mathbb{C})$ too. This means

$$\int_{\mathbb{R}} |f|^2 d\lambda \leq \|f\|_\infty \int_{\mathbb{R}} |f| d\lambda = \|f\|_\infty \|f\|_1 < \infty,$$

hence $f \in L^2(\mathbb{R}; \mathbb{C})$, and by the same logic $\widehat{f} \in L^2(\mathbb{R}; \mathbb{C})$.

(b) I think $f, f' \in L^1$ should imply f is bounded? Assuming so, by Hölder's inequality,

$$\|f\|_2^2 = \|f^2\|_1 \leq \|f\|_1 \|f\|_\infty < \infty,$$

so $f \in L^1 \cap L^2$. By Plancherel's theorem, $\|\widehat{f}\|_2 = \|f\|_2 < \infty$, so $\widehat{f} \in L^2$.

(c) By Axler's 11.54, $|\widehat{f}(t)| = 4\pi^2 t^2 |\widehat{f''}(t)|$ for all $t \in \mathbb{R}$. By Axler's 11.49, $\widehat{f}$ and $\widehat{f''}$ are both in L^∞ . Putting this together,

$$\int_{\mathbb{R}} |\widehat{f}| d\lambda = \int_{[-1,1]} |\widehat{f}| d\lambda + \int_{\mathbb{R} \setminus [-1,1]} |\widehat{f}| d\lambda \leq 2\|\widehat{f}\|_\infty + \int_{\mathbb{R} \setminus [-1,1]} \frac{|\widehat{f''}|}{4\pi^2 t^2} d\lambda < \infty,$$

where the integral of $|\widehat{f''}|/4\pi^2 t^2$ is finite because $\widehat{f''}$ is bounded; thus $\widehat{f} \in L^1$. □

Problem 5. (a) Let $f \in L^2(\mathbb{R}; \mathbb{C})$. Show that $\mathcal{F}^2 f(x) = f(-x)$ a.e. on $\mathbb{R}$.

(b) Suppose $f \in L^1(\mathbb{R}; \mathbb{C}) \cap L^2(\mathbb{R}; \mathbb{C})$ and $g(x) = \int_{\mathbb{R}} f(t) e^{2\pi i x t} d\lambda(t)$ for $x \in \mathbb{R}$. Show that $g(x) \in \mathbb{C}$ is well-defined for all $x \in \mathbb{R}$, $g \in L^2(\mathbb{R}; \mathbb{C})$, and $\mathcal{F}g = f$.

Solution. (a) Suppose $f \in L^1 \cap L^2$ and, for $y > 0$, let P_y be the Poisson kernel. By #3(c),

$$\|f * P_y\|_p \leq \|f\|_1 \|P_y\|_p < \infty$$

for all $p \in [1, \infty]$ because $P_y \in L^p$ for all $p \in [1, \infty]$, as shown in class, and $\widehat{f * P_y} = \widehat{f} * \widehat{P_y} \in L^p$ because it is the product of an L^∞ function with an L^p function. In particular, the previous sentence shows that $f * P_y, \mathcal{F}(f * P_y) \in L^1 \cap L^2$, so Fourier inversion implies that $\mathcal{F}^2(f * P_y)(x) = f * P_y(-x)$ a.e. Taking the limit in L^2 as $y \rightarrow 0^+$ and applying the result of #7(a) implies that $\mathcal{F}^2 f(x) = f(-x)$ a.e. for all $f \in L^1 \cap L^2$, which implies the result for all $f \in L^2$ because $L^1 \cap L^2$ is dense in L^2 and $\mathcal{F}: L^2 \rightarrow L^2$ is continuous.

(b) The integral defining g is well-defined because $f \in L^1$; explicitly,

$$|g(x)| \leq \int_{\mathbb{R}} |f(t) e^{2\pi i x t}| d\lambda(t) = \|f\|_1 < \infty.$$

Furthermore, $g(-x) = \widehat{\widehat{f}}(x) = \mathcal{F}f(x)$ for all $x \in \mathbb{R}$ by definition of the Fourier transform, so Plancherel's theorem implies $\|g\|_2 = \|f\|_2 < \infty$. Also, using (a), we get that for all $x \in \mathbb{R}$,

$$f(x) = \mathcal{F}^2 f(-x) = \mathcal{F}g(x). \quad \square$$

Problem 6. (a) Let $f, g \in L^2(\mathbb{R}; \mathbb{C})$. Show that $fg \in L^1(\mathbb{R}; \mathbb{C})$ and $\widehat{fg} = (\mathcal{F}f) * (\mathcal{F}g)$.

(b) Let $f, g \in L^2(\mathbb{R}; \mathbb{C})$ with $f * g \in L^2(\mathbb{R}; \mathbb{C})$. Show that $\mathcal{F}(f * g) = (\mathcal{F}f)(\mathcal{F}g)$ a.e. on $\mathbb{R}$.

Solution. (a) Consider the convolution map $- * -: L^2 \times L^2 \rightarrow L^\infty$, which is well-defined because $\|f * g\|_\infty \leq \|f\|_2 \|g\|_2$ by #3(c). For $f, g \in L^2$, each map $f * -: L^2 \rightarrow L^\infty$, $- * g: L^2 \rightarrow L^\infty$ is linear (because convolution is bilinear) and bounded (because $\|f * g\|_\infty \leq \|f\|_2 \|g\|_2$), so convolution is continuous in each component, whence it is easy to see that the convolution map on $L^2 \times L^2$ (with the product metric $d((f_1, g_1), (f_2, g_2)) = \|f_1 - f_2\|_2 + \|g_1 - g_2\|_2$) is continuous.

Since $\mathcal{F}: L^2 \rightarrow L^2$, $\widehat{\bullet}: L^1 \rightarrow L^\infty$, and the pointwise multiplication map $L^2 \times L^2 \rightarrow L^1$ are continuous (the second one by Riemann-Lebesgue and the latter by Hölder/Cauchy-Schwarz), it suffices to show that $(f, g) \mapsto (\mathcal{F}f) * (\mathcal{F}g)$ agrees with $(f, g) \mapsto \widehat{fg}$ on the dense subset $(L^1 \cap L^2) \times (L^1 \cap L^2) \subseteq L^2 \times L^2$. Indeed,

$$\mathcal{F}(\mathcal{F}f * \mathcal{F}g)(-x) = (\mathcal{F}^2 f \cdot \mathcal{F}^2 g)(-x) = f(x)g(x) \iff \mathcal{F}^2(\mathcal{F}f * \mathcal{F}g)(-x) = (\mathcal{F}f * \mathcal{F}g)(x) = \widehat{fg}(x),$$

for all $x \in \mathbb{R}$, using the result of (b) for the leftmost equality (the proof of (b) below doesn't rely on (a)).

(b) As shown in (a), convolution is continuous on $L^2 \times L^2$; thus by density it suffices to verify the given identity for $f, g \in L^1 \cap L^2$. In this case $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ by #3(c), so $\widehat{f * g}$ is well-defined, meaning

$$\mathcal{F}(f * g) = \widehat{f * g} = \widehat{fg} = (\mathcal{F}f)(\mathcal{F}g),$$

so we are done. $\square$

Problem 7. Let $f \in L^1(\mathbb{R}; \mathbb{C})$.

(a) For any $y > 0$, let P_y be the Poisson kernel of Axler's 11.68. Show that $P_y * f, \widehat{P_y * f} \in L^p(\mathbb{R}; \mathbb{C})$ for all $p \in [1, \infty]$, $P_y * f$ converges to f as $y \rightarrow 0^+$ with respect to $\|\bullet\|_p$ for all $p \in [1, \infty)$ such that $f \in L^p(\mathbb{R}; \mathbb{C})$, and $\widehat{P_y * f}$ converges to $\widehat{f}$ as $y \rightarrow 0^+$ with respect to $\|\bullet\|_p$ for all $p \in [1, \infty]$ such that $\widehat{f} \in L^p(\mathbb{R}; \mathbb{C})$.

(b) Suppose $\widehat{f} \in L^2(\mathbb{R}; \mathbb{C})$. Show that $f \in L^2(\mathbb{R}; \mathbb{C})$.

Solution. (a) Recall that $P_y, \widehat{P_y} \in L^p$ for all $p \in [1, \infty]$ and $y > 0$ (as shown in class). Using #3(c),

$$\|P_y * f\|_p \leq \|P_y\|_p \|f\|_1 < \infty,$$

so $P_y * f \in L^p$. Furthermore $\widehat{P_y * f} = \widehat{P_y} \widehat{f} \in L^p$ because it is the product of the L^p function $\widehat{P_y}$ with the bounded function $\widehat{f}$.

Convergence of $P_y * f$ to f when $f \in L^p$ for $p \in [1, \infty)$ is exactly shown in Axler's 11.74. To show that $\widehat{P_y * f}$ converges to $\widehat{f}$ as $y \rightarrow 0^+$ if $p < \infty$, we compute

$$\lim_{y \rightarrow 0^+} \|\widehat{f} - \widehat{P_y} \widehat{f}\|_p^p = \lim_{y \rightarrow 0^+} \int_{\mathbb{R}} |\widehat{f}|^p \cdot |1 - \widehat{P_y}|^p d\lambda = 0$$

with the last equality following from dominated convergence, where the integrand converges to and is uniformly bounded by $|\widehat{f}|^p$ as $y \rightarrow 0^+$. If $p = \infty$ then we note that $\widehat{P_y} = e^{-2\pi y|t|}$ converges uniformly to 1 as $y \rightarrow 0^+$, so, since $\widehat{f} \in L^\infty$ is bounded almost everywhere, $\widehat{P_y} \widehat{f}$ converges with respect to the L^∞ norm to $\widehat{f}$.

(b) I reduced this to showing that $\|f * P_y\|_2$ converges to $\|f\|_2$ as $y \rightarrow 0^+$ if $\|f\|_2 = \infty$ (trying to show that $f \notin L^2$ implies $\widehat{f} \notin L^2$) but I don't see how to get the convergence (likely missing some silly bound on $f * P_y$ or something). $\square$